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Whole-genome sequencing of presumptive MDR-TB isolates from a tertiary healthcare setting in Mumbai

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ABSTRACT

Objectives: Whole-genome sequencing (WGS) of *Mycobacterium tuberculosis* (MTB), proven to be a better alternative when compared with the combined sensitivity and specificity of all other modalities for diagnosis of tuberculosis (TB), aids epidemiological surveillance investigations by combining the current research with diagnostics. This study was conducted to identify and resolve operational challenges in performing WGS-based drug resistance testing (DRT) for MTB in a TB culture and drug susceptibility testing (DST) laboratory. Three critical, non-redundant steps for WGS-based DRT were tested: viz. DNA extraction, high-throughput paired-end next-generation sequencing (NGS), and genomic analysis pipeline for automated reporting of WGS-based DRT.

Methods: DNA was extracted from 100 liquid culture isolates on a mycobacterial growth indicator tube (MGIT) using DNEASY Ultraclean Microbial Kit (Qiagen, USA) as per the manufacturer's instructions. Illumina paired-end sequencing was performed. All analysis steps were automated using custom python scripts, requiring no intervention. Variant calling was performed as per the World Health Organization (WHO) technical guide.

Results: The number of cultures resistant to rifampicin, isoniazid, pyrazinamide, ethambutol, and streptomycin was 89, 88, 35, 67, and 73, respectively. Resistance to amikacin, kanamycin, and capreomycin was found in 15, 17, and 15 cultures, respectively. Seventy cultures were resistant to fluoroquinolones, four were resistant to ethionamide, and 12 were resistant to linezolid. Six cultures were resistant to only one of the 18 drugs tested. Seventy-five cultures were resistant to more than three anti-TB drugs. One culture was resistant to 13 of the 18 anti-TB drugs tested for this study. The maximum number of variants were observed in the *rpoB* gene ($n = 93$, 93%), wherein the Ser450Leu was the predominant mutation ($n = 68$, 73%). Ser315Thr was the most common variant ($n = 86$, 97%) that encoded resistance to isoniazid. The Lys43Arg variant encodes resistance to streptomycin and was the third most predominant variant ($n = 65$, 89%). In addition to the high levels of resistance observed in the dataset, we also observed a high proportion of Beijing strains ($n = 63$, 63%).

Conclusion: Compared with results from routine diagnostics based on the 'Guidelines on Programmatic Management of Drug-Resistant TB (PMDT) in India', none of the samples had DST available for all 18 drugs. This represents a gap in PMDT guidelines. The WGS-DRT must be considered as the primary DST method after a sample is flagged rifampicin-resistant by cartridge-based nucleic acid amplification testing (CBNAAT). With several research studies currently underway globally to identify novel variants associated with drug resistance and classify their minimum inhibitory coefficients, WGS-DRT presents a scalable technology that updates analytical pipelines, relegating the need for changing microbiological protocols.

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1. Introduction

Whole-genome sequencing (WGS) has emerged as a key technology focused on the detection of drug resistance (DR) of *Mycobacterium tuberculosis* (MTB). On this front, Public Health England was the first to recognise the power of this technology and, in 2017, routine WGS in clinical settings of the National Health Service was introduced. This introduction served as a primer for others to incorporate WGS into various public health facilities around the globe. Also in 2017, the World Health Organization (WHO) released the 'Technical guide on next-generation sequencing technologies for the detection of mutations associated with drug resistance in *Mycobacterium tuberculosis complex*', which was foreseen to overcome the technical and managerial challenges faced (especially by low-income countries with high tuberculosis (TB) burdens) in incorporating WGS into routine diagnostic laboratories.

In India, the 2017 Programmatic Management of Drug-Resistant TB (PMDT) [1] guidelines set by the Revised National Tuberculosis Control Programme (RNTCP) (now known as the National Tuberculosis Elimination Programme [NTEP]) suggest the use of rapid molecular-based methods, such as the nucleic acid amplification tests (NAATs) GeneXpert and Line Probe Assay, for the detection of tuberculosis and drug resistance. However, because of their sensitivity, the scope of these kits is currently limited [2]. Furthermore, they require validation through the gold-standard phenotypic DST test, Culture DST (CDST) [3–5], leading to diagnosis times of weeks to months and resulting in poorer patient outcomes. India, a high-burden country, accounts for 24% of the world's rifampicin-resistant (RIF^R)/multi-drug resistant (MDR) TB cases [6]. However, only every third patient who developed MDR-TB or RIF^R-TB was detected and reported [6]. Despite the near-complete coverage of DOTS (Directly Observed Therapy, Short-course), there is rising concern regarding an increasing incidence of drug-resistant tuberculosis (DR-TB) [7] that may be fueled by treatment delays and ongoing DR-TB transmission [7–11]. Implementing WGS, particularly in high-burden and resource-limited settings such as India, will aid in the prompt identification of drug resistance.

In this study, we demonstrate the use of WGS for *Mycobacterium tuberculosis* samples by assessing presumptive MDR-TB samples from one of the diagnostic referral labs to validate its necessity in a CDST laboratory within the NTEP. Furthermore, we present additional information derived from WGS data analysis, including lineage prevalence, MDR/extremely drug-resistant (XDR) TB surveillance, the co-occurrence of resistance, and distributions of resistance-causing mutations. This study also shows the scalability of implementing WGS nationally for the development of better facilities for TB care.

2. Materials and Methods

2.1. Study settings

The study was conducted at the TB culture and drug susceptibility testing laboratory (CDST) in Grant Government Medical College and Sir. J. J. Hospital (GGMC & JJ), Mumbai, India. The CDST laboratory is certified by the NTEP for solid culture and liquid DST for TB cultures. One hundred TB cultures, culture-positive on *Mycobacterium* Growth Indicator Tube (MGIT) and tested as presumptive MDR-TB samples in accordance with PMDT guidelines, were selected for WGS. Ethical approval was obtained from the institutional ethics committee of Sir J. J. Hospital. Patient identities were masked and their data barcoded before cultures were used for the study.

2.2. Sample preparation and DNA extraction

An initial enrichment process was carried out to maximise the amount of MTB DNA obtained. Stocked positive MGITs were selected, and a 1.5 mL culture was aliquoted in a sterile 2 mL screw-cap tube. Two such aliquots were made for each culture and subjected to heat-killing in a water bath at 95°C for 30 min. The aliquots were centrifuged at 10,000 xg for 5 min, or until a tight pellet was formed. The pellets were suspended in 100 to 150 µL of sterile phosphate buffer saline (PBS) with a pH of 7.4. The two aliquots were pulled together and further processed for DNA extraction using the Qiagen DNeasy UltraClean Microbial Kit (Cat. no. 12224, Qiagen, Germany) according to the manufacturer's instructions, with a minor modification of elution volume, which was changed to 50 µL.

The qualitative and quantitative examination of the extracted DNA was performed using the Qubit dsDNA HS Assay kit (ThermoFisher Scientific, Catalogue: Q32851, USA) on a Qubit Fluorimeter (ThermoFisher Scientific, Catalogue: Q33226, USA) and NanoDrop (ThermoFisher Scientific, USA). DNA with a concentration greater than 0.2 ng/µL and $A_{260/280} = 1.8$ were used for WGS.

2.3. Library preparation and sequencing

Library preparation was performed using the PlexWell WGS-24 Library Preparation protocol (Part No. WGS24). Approximately 10 ng of Qubit quantified genomic DNA was simultaneously tagged (fragmented and adaptor tagged) and uniquely barcoded with Index i7 using Sample-Barcoding (SB) Reaction Set-up provided in the PlexWell Kit. The uniquely barcoded index i7 DNA was pooled for further pool barcoding process using Index i5 barcode reaction setup, and then subjected to 12 cycles of FILL-AMP PCR (72 °C for 10 mins followed by denaturation at 95 °C for 3 min, cycling [98 °C for 30 sec, 64 °C for 15 sec, and 72 °C for 30 sec], and 72 °C for 3 min) to enrich the adapter-tagged barcoded fragments. The polymerase chain reaction (PCR) product was purified using HighPrep Beads (MAGBIO genomics). Quantification and size distribution of the prepared pooled library was determined using Qubit fluorimeter and Agilent 2200 TapeStation, respectively, as per the manufacturer's instructions.

2.4. Adapter details

Universal Adapter
 5'-AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGC
 TCTTCCGATCT-3'
 Adapter, Index
 5'-GATCGGAAGAGCACACGTCTGAACTCCAGTCAC[INDEX]ATCTCGT
 ATGCCGTCTTCTGCTTG-3'

The Illumina-compatible sequencing library pools had an average fragment length of ~630 bp and were suitable for 150 × 2 paired-end Illumina sequencing.

2.5. Genomic analysis of drug resistance

Quality and depth of sequencing were analysed using FastQC [12]. Raw reads were subjected to adapter trimming using Trimmomatic [13] and processed using MTBSeq pipeline v. 1.0.3 for mapping of the complete genome of *Mycobacterium tuberculosis* H37Rv (NC_000962.3) and variant calling. Briefly, the trimmed reads were deduplicated, followed by genome mapping [14,15] and variant calling [16]. Low-frequency variants were called by the pipeline based on the following criteria: a minimum of two alleles to be detected on the forward and reverse reads individually; at least four reads calling the allele; Phred score of ≥20;

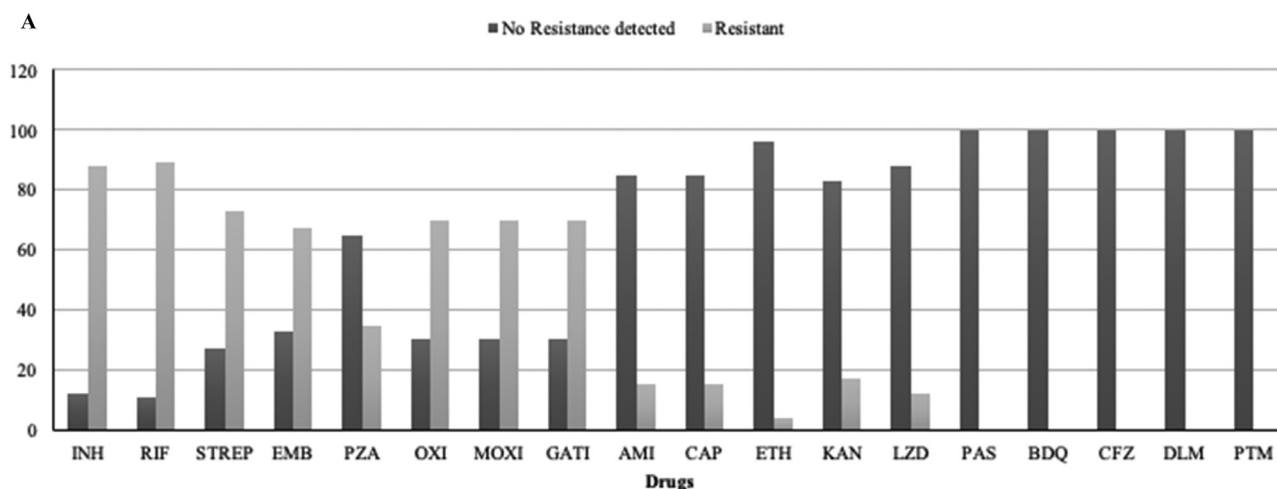


Fig. 1. (A) Distribution of samples from the cohort across all the antibiotics. Black bars represent the number of samples in which no resistance-imparting mutations were detected for a respective antibiotic; grey bars represent the number of samples for which resistance-imparting mutations were detected for the respective antibiotic. (B) Classification of drug resistance among the samples into DS-TB, mono-resistance to INH/RIF, MDR, and XDR. *Includes one sample that was detected as resistant to only EMB.

and the minimum allele- frequency of 20. In addition to the resistance-mediating mutations included in MTBSeq [17] (isoniazid [INH], rifampicin [RIF], streptomycin [STREP], ethambutol [EMB], pyrazinamide [PZA], fluoroquinolones [FQ], ofloxacin [OXI], moxifloxacin [MOXI], gatifloxacin [GATI], amikacin [AMI], capreomycin [CAP], ethionamide [ETH], kanamycin [KAN], linezolid [LZD], clofazimine [CFZ], and para-aminosalicylic acid [PAS]), these variants were screened against a comprehensive catalogue for identification of lineage and resistance. The catalogue was expanded further to include recently identified variants responsible for imparting resistance against bedaquiline (BDQ), delamanid (DLM), and pretomanid (PTM) [14–24].

Custom R scripts were used to identify the genomic DRT and extract variants to further identify the co-occurrence of mutations in the samples. Variants that occurred in five or more samples were selected to see co-occurrence patterns in those variants.

2.6. Phylogenetic analysis

MEGAX platform was used for phylogenetic analysis of the samples [18,19]. Single nucleotide polymorphisms (SNPs) identified from the samples were concatenated to generate individual FASTA sequences for each sample. Multiple sequence alignment was performed using ClustalW, which was used for the phylogenetic tree reconstruction by the Maximum-likelihood method with 1000 bootstrap replicates.

3. Results and discussion

The number of reads in the samples ranged from 0.66 to 176 million. The average number of reads that mapped to the reference (H37Rv) was 94.76% (1.55% – 99.41%) (Supplementary Fig. S1). Five samples mapped to less than 50% of the reference. However, we were still able to predict resistance in these samples; hence they were included in this study. For the variants screened for resistance prediction, the average variant depth for all samples ranged from 30.84 to 150.85.

3.1. General prevalence of drug resistance

Ninety-four isolates were found to be resistant to at least one of 18 antibiotics. The number of isolates resistant to first-line drugs

RIF, INH, PZA, EMB, and STREP was 89, 88, 35, 67, and 73, respectively (Fig. 1A). Seventy samples were resistant to FQ, and 17 samples were resistant to at least one of the second-line injectables. Only four samples were resistant to ethionamide. One sample was resistant to 13 of the 18 drugs tested. None of the samples were resistant to PAS, CFZ, or the three newly introduced drugs in the NTEP protocol, BDQ, DLM, and PTM. Less than 50% of reads mapped to the reference for five samples; among these, three showed the presence of resistance-conferring mutations detected for at least one of the drugs (Data S1).

Since the cultures used in this study were sampled from patients diagnosed and treated according to the 2017 PMDT guidelines, the analysis and inferences drawn here are based on the same guidelines. According to the 2017 PMDT guidelines, MDR-TB is defined as MTB-resistant to both RIF and INH; additionally, resistance to any FQ and at least one of the second-line injectables is defined as XDR-TB. These definitions were considered for the classification of the samples in the study. Of the 100 presumptive MDR-TB samples, 70 were MDR-TB and 14 were XDR-TB (Fig. 1B). Among 70 MDR-TB samples, 54 were resistant to FQ, and 3 were resistant to second-line injectables (Data S1). Notably, nine samples had resistance to either RIF or INH and, hence, could not be classified as MDR-TB (Data S1). Collectively, 73 samples were resistant to either FQ and/or second-line injectables, as well as rifampicin and isoniazid. Twelve of the 84 MDR-TB/XDR-TB samples were resistant to linezolid. Surprisingly, six of the suspected MDR-TB samples did not have any resistance-conferring mutations, and one sample was resistant to only ethambutol (Fig. 1B).

3.2. Predominant mutations

Seventy-three unique mutations were identified in the study dataset, which was used to predict drug resistance (Data S2). Sixty-eight of the RIF^R samples had the C1349T mutation in the *rpoB* gene, resulting in Ser450Leu substitution, while 86 of the INH^R samples had a mutation in the *katG* gene, resulting in Ser315Thr substitution (Fig. 2, Supplementary Fig. S2). Thirteen of 17 samples resistant to at least one second-line injectable had a point mutation at position 1401 in the *rrs* gene (Fig. 2, Supplementary Fig. S2). Most ($n = 68/70$) of the FQ resistance resulted from mutations in the *gyrA* gene. The predominant mutation in these samples was Asp94Gly ($n = 34$), followed by Ala90Val ($n = 18$)

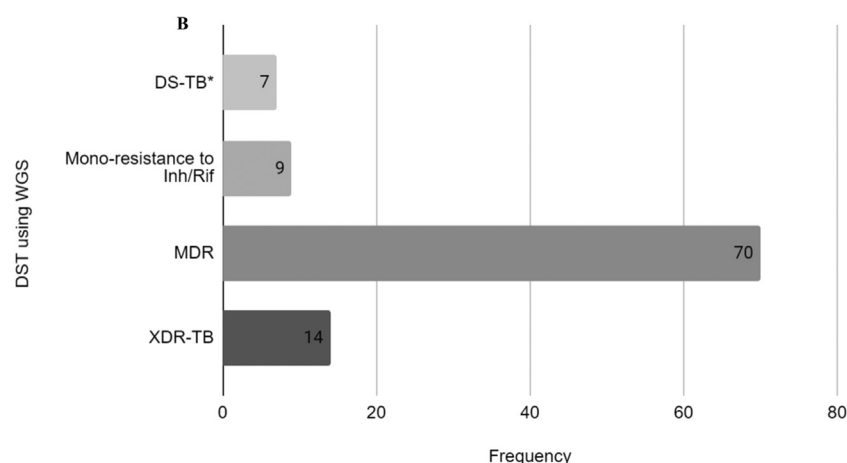


Fig. 2. (A) Predominant mutations among the samples across the antibiotics. (B) Frequency of the mutations across the respective antibiotics.

Table 1

Two variants of the *rpoB* gene, Leu430Pro and Asp435Tyr, show similar co-occurrence of the mutations across other genes viz. *rrs*, *pncA*, *embB*, *gyrA*, *rpoB*, and *rpoC*

Gene Name	A.A (Gene position for non-A.A. coding region)	Genome position	Leu430Pro of <i>rpoB</i>				Asp435Tyr of <i>rpoB</i>			
			Samples containing both mutation	Samples containing only given mutation	Samples containing only Leu430Pro	Samples not having either of the mutations	Samples containing both mutation	Samples containing only given mutation	Samples containing only Asp435Tyr	Samples not having either of the mutations
<i>rrs</i>	1401	1473246A>G	0	13	7	74	0	13	7	74
<i>pncA</i>	Gly132Ala	2288847C>G	0	7	7	80	0	7	7	80
<i>pncA</i>	Leu27Pro	2289162A>G	0	9	7	78	0	9	7	78
<i>pncA</i>	-11 intergenic region	2289252T>C	0	5	7	82	0	5	7	82
<i>embB</i>	Gly406Ser	4247729G>A	0	5	7	82	0	5	7	82
<i>gyrA</i>	Ala90Val	7570C>T	1	17	6	70	1	17	6	70
<i>gyrA</i>	Asp94Tyr	7581G>T	0	5	7	82	0	5	7	82
<i>rpoB</i>	Ser450Leu	761155C>T	0	68	7	19	0	68	7	19
<i>rplC</i>	Cys154Arg	801268T>C	0	10	7	77	0	10	7	77

(Fig. 2, Supplementary Fig. S2). The predominant mutation conferring resistance to streptomycin was A128G in the *rpsL* gene resulting in a Lys43Arg substitution (Fig. 2, Supplementary Fig. S2). Among the linezolid-resistant samples, the Cys154Arg mutation in the *rplC* gene was predominant ($n = 10$) (Data S2).

3.3. Co-occurrence of mutations leading to resistance among drugs

Eighteen unique mutations were checked for co-occurrence in the dataset. Seven samples having resistance to PZA had a C395G mutation in the *pncA* gene resulting in a Gly132Ala substitution. Our co-occurrence analysis of mutations imparting resistance showed that all seven samples also had Gln497Arg in *embB*, Asp94Gly in *gyrA*, Ser450Leu in *rpoB*, and Lys43Arg in *rpsL*, imparting EMB, FQ, RIF, and streptomycin resistance, respectively. These samples also showed the presence of Ser315Thr in the *katG* gene and an intergenic mutation at the -15 position of the *fabG1* gene, imparting INH resistance (Data S1). Because of the A1401G mutation in the *rrs* gene, thirteen samples having resistance to second-line injectables (AMI, KAN, and CAP) also had the Ser315Thr mutation in *katG*, imparting resistance to INH. Nine samples having PZA resistance had an A80G mutation in *pncA*, resulting in Leu27Pro substitution. All these samples had Ser315Thr in *katG*, Met306Val in *embB*, Ser450Leu in *rpoB*, and Lys43Arg in *rpsL*, imparting INH, EMB, RIF, and STREP resistance, respectively. Five samples showing a T-15C mutation at the intergenic region of the *pncA* gene, resulting in PZA resistance, had Ser450Leu and Ser315Thr mutations imparting resistance to RIF and INH, respectively. Furthermore, 34 FQ-resistant samples with A281G in

gyrA resulting in Asp94Gly had Ser315Thr in *katG*, resulting in INH resistance (Data S1). Interestingly, we also observed a pattern in the presence of Leu430Pro and Asp435Tyr in *rpoB*. None of the samples had these mutations; however, nine other mutations from different genes showed similar co-occurrence patterns (Table 1).

3.4. Lineage identification

Lineage classification was based on phylogenetic SNPs characterised by Coll et al. in 2014 [27]. One-third (63%) belonged to lineage 2 (61-Beijing and 2-East Asian), 15 belonged to lineage 3 (Delhi-CAS), 15 to lineage 4 (6-European American, 1-Haarlem, 2-LAM, and 1-Mainly T), and 6 from lineage 1 (East African Indian Ocean) (Fig. 3A). Thirteen of the fourteen XDR-TB samples belonged to the Beijing lineage (Fig. 3B, Data S1).

3.5. Cluster identification and phylogenetic analysis

Genomic clusters (GCs) within the samples were identified based on their pairwise distances being not more than 12 SNPs. It was found that four distinct GCs could be identified from the cohort and all the members were from the Beijing lineage (Table 1). The concatenated SNP alignment was used for phylogenetic analysis. Distinct clusters of MTB lineages determined by the phylogenetic markers were observed. The GCs identified using SNP distance formed identifiable clades (Fig. 4). All members of GC1 were MDR-TB. Two samples (JJ-31 and JJ-58) that belonged to this cluster probably diverged early from the common ancestral node

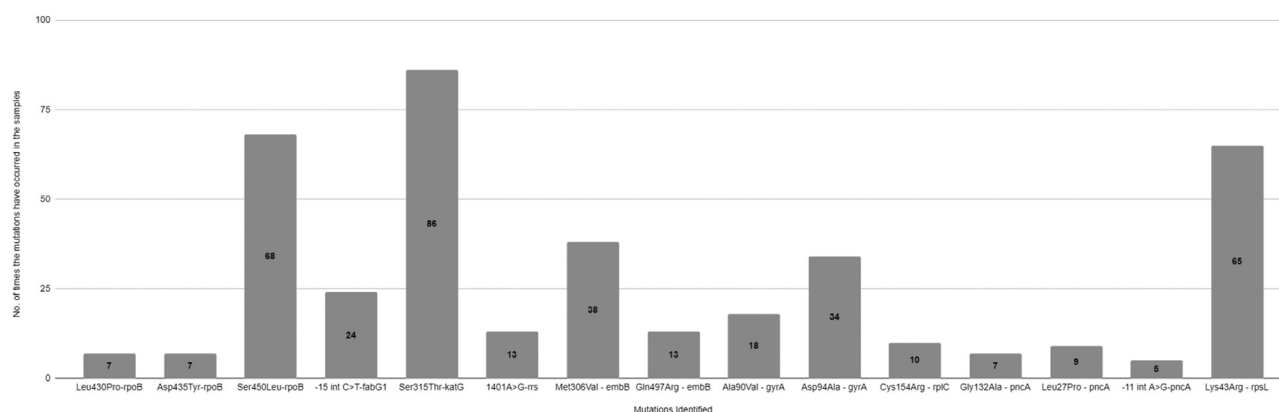


Fig. 3. (A) Distribution of samples into Coll lineages showing most of the samples belonging to the Beijing lineage (Blue). (B) Distribution of all the lineages into resistance patterns: DS-TB, mono-resistance to INH/RIF, MDR-TB, and XDR-TB.

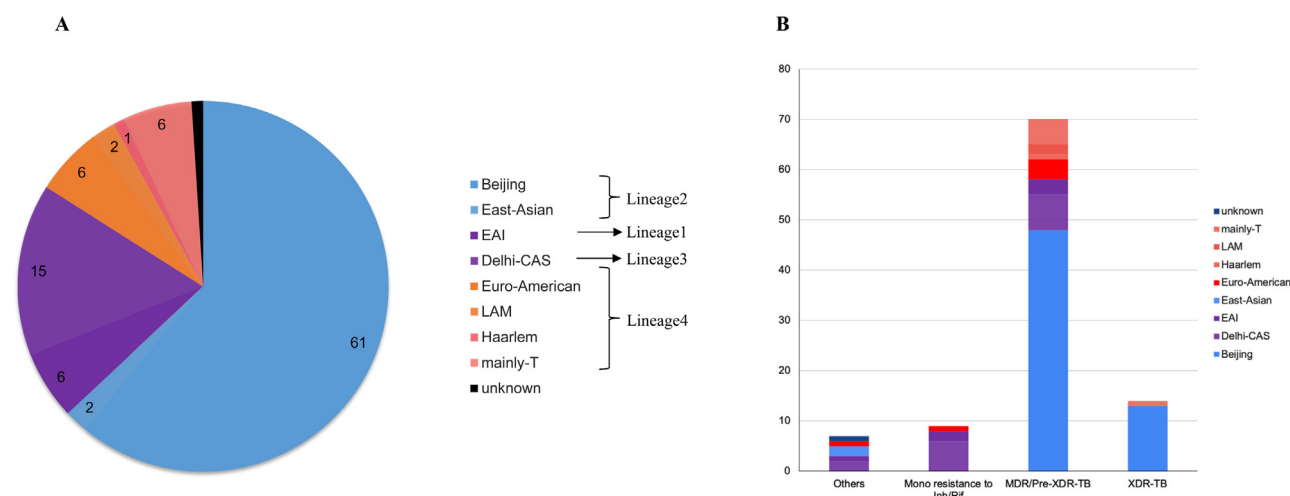


Fig. 4. Phylogenetic tree representing the genomic clusters among the samples. Blue, Beijing lineage; GC, genomic cluster.

(Fig. 4). A similar phenomenon was observed for GC2, where samples JJ-125 and JJ-23 diverged early from the cluster.

Interestingly, one of the samples (JJ-3) in GC2 was observed to have evolved into an XDR-TB case (Fig. 4). The clade consisting of all the members of GC3 had three members and might have diverged significantly to be included in the cluster. Sample JJ-70 might have diverged early from the GC3 members from their ancestral node. However, samples JJ-87 and JJ-110 might have diverged further from the cluster. Three samples from GC3, JJ-140, JJ-115, and JJ-80, identified as XDR-TB, and JJ-140 might have diverged early from the sub-clade. However, samples JJ-115 and JJ-80 might have accumulated mutations leading to their divergence as XDR-TB from the other MDR-TB sub-clade members (Fig. 4). GC4, consisting of 25 samples, had 7 XDR-TB samples. Among the XDR-TB samples, only sample JJ-59 had diverged early from the sub-clade and further evolved as an XDR-TB. However, other samples (JJ-41, JJ-139, JJ-149, JJ-37, JJ-92, and JJ-94) had accumulated resistant mutations later to form an inner node (Fig. 4).

3.6. Discussion

As part of an operations research study, we attempted to understand the need and feasibility of performing WGS-based DRT in presumptive MDR-TB cases in a tertiary care hospital under the RNTCP/NTEP. The study was conducted as a blinded SNP-based genotypic drug resistance determination to support the reliability and applicability of WGS in public health care settings. The initial and most crucial step towards this is the integrity of the se-

quencing data, which is determined by the depth of coverage and the percentage of mapped raw reads. Also, for determining SNPs, raw reads can provide greater resolution because of the quantifiable depth of coverage and proportion of mixed alleles compared to using a genome assembly in which all coverage at a given locus is collapsed into a single base call [20]. Hence, we performed a mapped raw read analysis and determined the proportion of mixed alleles. Of 100 samples, 5 samples have less than 50% of reads mapped to the reference sequence and, hence, have considerably low depth at the genome positions corresponding to resistance through variance. However, we identified variants in these samples based on the presence of at least two alleles in forward and reverse reads individually and at least four reads calling the allele. This suggests that genotypic drug resistance determination using WGS can be done on samples of sputum, which has a limitation of percentage mapped reads because of reads other than MTB. However, this study has a limitation of the number of samples; hence, further evaluation with a larger sample cohort and determination of screening parameters for qualifying the variant called for predicting resistance must be established.

The predominant mutation in the *katG* gene resulting in INH resistance is Ser315Thr, which was also found to be a predominant mutation in a study conducted in Russia [21]; it has been associated with high-level resistance (MIC > 1 µg/mL) [22]. Furthermore, it has been reported that a high dose cannot treat a high level of INH resistance due to this mutation [23]. However, according to the current PMDT guidelines, all samples which are RIF-resistant and FQ- and second-line injectable drug (SLID)-sensitive are given

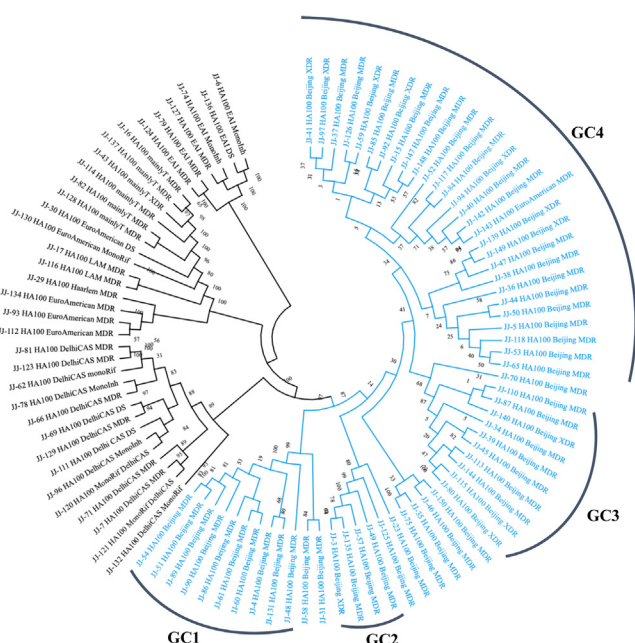


Fig. 5. Schematic representation of the 2021 PMDT guidelines. FL-LPA, first-line probe assay; SL-LPA, second-line probe assay; LC-DST, liquid culture DST; Z, pyrazinamide; Bdq, bedaquiline; Cfz, clofazimine; Mfx, moxifloxacin; Lzd, linezolid; Dlm, delamanid; H, isoniazid; FQ, fluoroquinolones; Rif, rifampicin.

a shorter MDR-TB regimen consisting of high-dose INH (along with other drugs) during the intensive period. INH-resistant TB is much more common than RIF-resistant TB. We suggest including DST for INH along with RIF at the initial point of diagnosis as part of the Universal DST for TB patients, followed continuous monitoring for any development of INH resistance among patients treated with INH.

Mutation at the 1401 position in the *rrs* gene, imparting resistance to at least one of the SLIDs, is the predominant mutation in the current cohort; it has been reported as the most common mutation in India, China, and Pakistan [24–27]. A high prevalence of FQ resistance was observed in the cohort, consistent with growing FQ resistance in Mumbai [28]. Interestingly, most of the mutations found were situated in the *gyrA* gene. This is consistent with the previous study, which reported the prevalence of mutations at *gyrA* codon 88 to 94 and the *gyrB* G512R mutation to be higher in Beijing and Indo-oceanic strains [29]. This suggests a close-to-saturation point reached for protection; judicious use of this broad-spectrum antibiotic should be followed [30].

According to the current NTEP guidelines, all presumptive and diagnosed TB patients must undergo a determination of RIF resistance (Fig. 5). RIF-susceptible patients will be subjected to first-line treatment (RIF, INH, EMB, and PZA) [1]. RIF-resistant patients will be subjected to the shorter MDR-TB regimen (MOXi^{high}, KAN, ETH, CFZ, PZA, INH^{high}, and EMB) [1]. In this study, a combination of INH monoresistance and no resistance to rifampicin was detected in five samples. While the RNTCP guidelines currently employ the shorter MDR-TB regimen consisting of high INH, these patients should be monitored for their reaction to INH. Conversely, we found four samples that were RIF-resistant with no resistance to INH. We believe that assessing INH along with RIF may solve these issues and give an added advantage of using the appropriate drug combinations during treatment.

Our lineage identification suggests that most presumptive MDR-TB samples belong to the Beijing lineage, consistent with the previous study. It was also found that 92–86% of the XDR-TB cases belonged to the Beijing lineage, suggesting the necessity of main-

taining surveillance on the transmission of this strain. It has been reported that strains belonging to the Beijing lineage have an increased ability to cause disease and spread within the geographical area [31]. Also, transmission analysis of TB is challenging in areas with large numbers of migrants, and standard genotypic methods fail to provide reliable transmission and epidemiological surveillance in such areas [32]. However, genomic epidemiology established through WGS has greatly increased our capacity to identify genomic clusters depicting transmission [33–36]. Determination of the number of SNPs between the patient samples facilitates sub-clustering of the samples into smaller GCs from the molecular method based on already identified larger clusters [34]. These GCs are reported to be more consistent with the geographical distribution of samples with epidemiological links between them [34]. We could identify four such GCs among samples belonging to the Beijing lineage. Identifying complex transmission patterns for changing socio-epidemiological features due to migration can help curb transmission at an early stage. To design and target SNP-based strain-specific PCR markers, WGS analysis of the samples is necessary. It should be deployed as a routine method of diagnosis and epidemiological study.

Of 100 samples, 73 would typically be subjected to liquid culture DST testing under the current PMDT guidelines. These were MDR-TB samples with resistance to either FQ and/or second-line injectables. Implementing WGS for TB instead of LC-DST at this stage would reduce the turn-around time from 3 weeks to 1 week and increase drug testing capacity to 18 drugs (Fig. 5). This would also result in faster implementation of DST for newly introduced drugs by real-time monitoring of their resistance-associated genes. WGS is also useful in getting a retrospective view of the clinical data to identify new mutations for drug resistance. Along with the success of the previously implemented Public-Private Interface Agency (PPIA) and the ongoing Joint Effort for Elimination of TB (JEET) program, the introduction of WGS will bring a one-stop solution that may help reduce transmission by lessening patient movement and decreasing the duration of the diagnostic pathway [37]. One of the positive findings of this study was that no mutations for BDQ, DLM, CFZ, and the newly introduced drug, PTM, were detected. This supports the utility of these drugs in MDR-TB patients who show resistance to more traditional drug regimens. One of the major drawbacks of this study was the low sample size, which was not sufficient for us to confirm or validate these findings. Further research is required. Additionally, there is a lack of metadata to verify the certainty of transmission analysis.

It has been more than a decade since the potential of WGS in the field of TB diagnosis and clinical decisions in public health settings was first demonstrated. However, MTB WGS remains non-implementable in various high TB burden countries, like India. One of the major hurdles in implementation is the strategic placement of WGS within the public health structure as defined by the PMDT guidelines. Through this study, we have shown the operational workflow by replacing phenotypic DST, the current gold standard of TB drug susceptibility testing, with WGS. Implementing WGS may reduce operational turnaround time and increase the precision of diagnosis with exponential scalability. The pre-XDR and XDR TB resistance classification criteria published by WHO in 2019 [38] advocates for the inclusion of LZD and BDQ in the group A drugs as well as FQ in the regimen. The resistance profiles for these drugs currently cannot be determined by existing molecular tests, such as line probe assay.

Furthermore, various new drugs, such as PTM, are on the verge of being included in the regimen because of their potential to treat TB. Monitoring resistance to these new drugs is crucial to avoid acquired resistance and prevent future outbreaks due to transmission. With robust quality control, precision analysis, and simplified interpretation, WGS holds great potential for achieving the Indian

TB National Strategic Plan's (NSP) 2017 to 2025 goal of eliminating TB in India.

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Competing interests

AZ, SS, and AC are employees of HaystackAnalytics Pvt. Ltd, which provides commercial genomics services. KK is a Founder at HaystackAnalytics and receives no financial support from Haystack Analysis. Other authors declare no competing interests.

Ethical approval

The project was approved by Institutional Ethics Committee, Grant Government Medical College, and Sir. J. J. Hospital (GGMC & JJ) of Mumbai, India.

Author contributions

AC, KK and AJ conceptualized the study. AZ performed genome sequencing. SS and AZ performed bioinformatics analysis. NH and AJ conducted the sample selection for the study and provided metadata for analysis.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jgar.2022.10.004](https://doi.org/10.1016/j.jgar.2022.10.004).

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